Gabriel Ona

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Master of Applied Science, University of Toronto
Portfolio/LinkedIn/GitHub
Edward S. Rogers Sr. Electrical and Computer Engineering

Research

Hardware/software co-design engineer specializing in large-scale HPC and AI systems, focused on the interconnect and communication bottlenecks that limit scaling of distributed AI workloads. Currently building FPGA/ASIC in-network accelerators and simulation infrastructure to characterize and mitigate these bottlenecks in MoE inference at rack- and datacenter-scale.

Education

- **University of Toronto** — Ontario, Canada
  Master of Applied Science - Computer Engineering — *Sep. 2025–Present*
  Advisor: Prof. Paul Chow
- **Universidad San Francisco de Quito** — Quito, Ecuador
  Bachelor of Science - Electrical Engineering — *Sep. 2020–Jul. 2025*

Experience

- **University of Toronto** — Ontario, Canada
  Summer Internships — *May 2024–Sep. 2024*
  - **High-Performance Reconfigurable Computing (HPRC)**:
    Ported a 100 Gbps FPGA-based Intrusion Detection and Prevention System (IDS/IPS) from Intel Stratix 10 to AMD Alveo U280 platforms.
  - **Space and Terrestrial Autonomous Robotic Systems (STARS)**:
    Designed a 3D Gaussian Splatting (3DGS) pipeline for non-fixed eye-to-hand calibration of robotic manipulators.
- **Synopsys Inc.** — Remote, Chile
  Bachelor's Co-op — *Nov. 2023–Nov. 2024*
  - **ASIC Design and Layout**:
    Designed a 64-bit dual-mode logic (DML) ripple-carry adder with critical path delay error detection and correction (EDAC) in a TSMC 14nm FinFET process node.
- **National Polytechnic School of Ecuador** — Quito, Ecuador
  Research Assistant — *Jan. 2020–May 2024*
  - **Aeronautics and Thermofluids Applications (ATA)**:
    Developed a ROS-based simulation pipeline to test custom UAVs under extreme weather conditions for Andean wetland monitoring.

Publications

- **Journal**: Valencia, E., Toapanta, F., Ona, G. et al. *An Open-source UAV Digital Twin Framework: A Case Study on Remote Sensing in the Andean Mountains.* Journal of Intelligent & Robotic Systems, 111, 71 (2025). doi:10.1007/s10846-025-02276-7

Projects

- **In-Network Acceleration for Distributed MoE Inference (Thesis Research)** — *July '26*
  - Built a discrete-event simulator modeling distributed MoE inference across thousands of GPUs.
  - Modeled multiple network fabrics (NVLink/NVSwitch, PCIe, Ethernet) and collective scheduling policies over configurable rack-scale topologies.
  - Benchmarked FPGA SmartNIC- and ASIC-based in-network acceleration at inter- and intra-rack boundaries.
  - Validated against DeepSeek-V4-Pro under stress conditions, achieving up to 6.7× speedup over a BSP baseline (combined ASIC and FPGA) and 1.7× with FPGA acceleration alone.
- **Apache Spark vs. Flink Benchmarking for Distributed Data Processing** — *April '26*
  - Built an industry-style benchmarking framework evaluating latency, throughput, scalability, and fault tolerance across an automated, configurable multi-node cluster.
  - Integrated a streaming testbed with Kafka-/SQL-based ingestion, NFS checkpointing, and Nexmark workloads.
  - Found Flink's exactly-once semantics improve latency but reduce throughput and slow failure recovery, compared to Spark's cheaper micro-batch-based recovery.
- **CNN-Based Informed Steering for RRT*** — *April '26*
  - Replaced RRT*'s heuristic-based steering with a ResNet-inspired CNN trained on three-channel (map, agent position, goal position) inputs, benchmarked against a pre-trained CNN with position-embedding injection.
  - Used distributed training with PyTorch DDP, learning rate scheduling, and synthetic dataset generation, improving prediction accuracy by 33%.

- **VLSI CAD Flow: Technology Mapping, Partitioning, Placement, and Routing** *September '25*
  - Built a C++ framework converting intermediate VLSI graph representations into low-level bitstreams across multiple benchmark circuits.
  - Integrated hypergraph partitioning, flow-based analytical placement, ABC-based technology mapping with custom standard cells, and maze routing.
- **FPGA-Based AI Accelerator with Reconfigurable Systolic Array Architecture** *January '25*
  - Designed a weight-stationary systolic array for general matrix multiplication (GEMM) from scratch, with a NoC-based topology for weight distribution and AXI-Lite/AXI-Stream control and data interfaces.
  - Parameterized all modules for reusable, scalable IP, scaling to a 256×256 array comparable to industry-grade accelerators.

Teaching

- **University of Toronto** Ontario, Canada
  Teaching Assistant *Jan. 2026–May 2026*
  - **ECE243 Computer Organization**: Prof. Vaughn Betz & Prof. Jonathan Rose
  - **ECE342 Computer Hardware**: Prof. Karthik Ganesan & Prof. Parinaz Naseri
- **Universidad San Francisco de Quito** Quito, Ecuador
  Teaching Assistant *Sep. 2024–May 2025*
  - **IEE2003 Signals and Systems**: Prof. Luis Miguel Procel
  - **IEE3200 Semiconductors**: Prof. Luis Miguel Procel

Honors and Awards

- **USFQ James Clerk Maxwell Full Tuition Scholarship** Quito, Ecuador
  Engineering Scholarship *2020–2025*
- **IEEE Distinguished Student Diploma** Quito, Ecuador
  Electrical Engineering Department Diploma *2023*
- **XXXVI Albert Einstein National Physics Contest** Quito, Ecuador
  Physics Bronze Medal *2019*
- **Infomatrix Science and Technology World Final** Bucharest, Romania
  Innovation Silver Medal *2019*
- **Solacyt Latin American Innovation and Technology Contest** Quito, Ecuador
  Innovation Gold Medal *2019*
- **XV Ecuadorian National Mathematics Olympiad** Quito, Ecuador
  Mathematics Bronze Medal *2018*

Volunteer Experience

- **Student Volunteer, IEEE International Solid-State Circuits Conference (ISSCC)** San Francisco, CA
  University of Toronto – Saratoga Group *Feb. 2026*
- **Founder & President, IEEE Aerospace and Electronic Systems Society (AESS)** Quito, Ecuador
  USFQ Student Branch Chapter *2023–2025*

Skills Summary

- **Programming Languages**: C/C++, Python, Bash, SQL, CUDA, Verilog/SystemVerilog, HLS, Assembly, TypeScript.
- **Computer Architecture & Hardware Design**: ASIC/FPGA design flows, RTL simulation & verification, Intel Quartus, Xilinx Vivado/Vitis, Synopsys Design Compiler/ICC2, Cadence OrCAD Capture/Allegro.
- **HPC & GPU Optimization**: Roofline Analysis, Discrete Event Simulators, NVIDIA Nsight (Systems/Compute), NCCL, OpenMPI, PyTorch (DDP), LLVM.
- **Networking & In-Network Systems**: NVLink, PCIe, Ethernet, FPGAs as SmartNICs, NoC, AXI Interfaces, UART/I2C/SPI/USB, BIND9, NGINX, Tailscale.
- **Cloud & Distributed Systems**: AWS, Terraform, Ansible, Docker/Compose, Kubernetes, Databricks, Apache Spark/Flink/Kafka/Zookeeper, Keycloak, OpenBao.
- **Spoken Languages**: Spanish (Native), English (C1), French (B2).